Magnitude, Not Shape: A Survivorship-Free, Significance-Tested Evaluation of Downside-Risk Metrics

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ABSTRACT

Every few years a new downside- or tail-risk metric is proposed to improve on plain return volatility, always on the same premise: the *shape* of the loss distribution — its asymmetry, tail heaviness, or co-crash structure — carries information that symmetric volatility misses. We test that premise as adversarially as the data allow. On a survivorship-free (a crypto panel retaining delisted coins, plus equity), multi-regime, multiple-testing-corrected evaluation of ten metrics across five test beds, no metric — published or machine-discovered — reliably beats volatility at forecasting an asset's forward maximum drawdown. The anatomy of that failure is a clean law: *magnitude, not shape.* The partial rank correlation of Value-at-Risk with forward drawdown, controlling for volatility, survives Benjamini–Hochberg correction on all five beds (up to 0.19), while every tail-*shape* metric adds nothing or subtracts. We explain why volatility is hard to beat (it is persistent, though a predictive mediation test shows not a sufficient statistic), show the tiny magnitude edge is not economically free (a cost-aware backtest and a pre-registered composite add nothing over volatility except in fat-tailed crypto), and demonstrate that an autonomous large-language-model discovery loop only inflates validation while its locked-test score plateaus at volatility — a data-snooping caution as agentic research makes signal search nearly free. Volatility is not glamorous, but on a level field it is very hard to beat.

CCS CONCEPTS

• **Computing methodologies → Machine learning**; • **Applied computing → Economics**.

KEYWORDS

downside risk, tail risk, drawdown, volatility, survivorship bias, multiple testing, data snooping, LLM agents

1 INTRODUCTION

The market's dangers are asymmetric: prices fall faster than they rise, and it is the fall that ruins investors. From this intuition a large literature has grown, proposing metrics that look past symmetric volatility to the *shape* of the loss distribution — lower partial moments and the Sortino ratio [6, 13, 19], the partial-moment "predictive" ratio we originally set out to audit [20], left-tail Value-at-Risk and expected shortfall [3, 18], realized semibetas [8], downside beta [2, 16], lower-tail dependence [9], disappointment-aversion risk [12], and tail-exponent factors [15]. Each rests on the same premise: that asymmetry or tail structure carries information volatility misses.

We test that premise on the fairest field we can build. Three disciplines the source literature usually omits turn out to matter enormously: a *survivorship-free* universe (so the assets that actually died are present), a *locked* out-of-sample holdout (so search cannot leak into the verdict), and *multiple-testing correction* (so an evaluation of dozens of comparisons does not manufacture winners). We evaluate ten metrics as forecasters of 90-day forward maximum drawdown across five cross-sectional test beds spanning calm, bull, and crash regimes in crypto and equity, with per-date rank-correlation skill, paired block-bootstrap confidence intervals, and Benjamini–Hochberg (BH) correction.

The headline is negative and, in hindsight, unsurprising: no metric reliably beats plain volatility. The contribution is the *anatomy* of that result and the honesty of the battery behind it. (1) We establish a clean empirical law — *magnitude, not shape*: the only signal that survives correction beyond volatility is the *size* of tail losses (VaR/ES/downside deviation), not their asymmetry (§5). (2) We explain why volatility is hard to beat — it is persistent — while honestly bounding that explanation: a predictive mediation test shows forward volatility is *not* a sufficient statistic (§6). (3) We ask whether the residual magnitude edge is economically real: a cost-aware backtest and a pre-registered composite show it is not, except in fat-tailed crypto (§7). (4) We show an autonomous large-language-model (LLM) discovery loop only overfits validation — a data-snooping caution for AI-driven research (§8). (5) We close with the one thing volatility is genuinely good for: predicting outright blow-ups (§9). Every robustness, economic, and discovery experiment is retained; supporting detail is in the appendices.

2 RELATED WORK

Downside-risk metrics. Lower partial moments generalize semivariance [6, 13] and underpin the Sortino ratio [19] and the partial-moment ratio we audit [20]. Tail-focused measures include left-tail VaR/ES [3, 18], realized semibetas [8], downside beta [2, 16], lower-tail dependence [9], disappointment-aversion risk [12], and tail-exponent factors [15]. Most were built to *price* the cross-section of returns, not to forecast a single asset's drawdown; we repurpose them and scope our conclusions accordingly. Separately, a drawdown-specific literature studies the maximum-drawdown statistic directly — its distribution under a Brownian model [17] and its use as a portfolio-optimization objective [10]. We borrow its *target* (forward maximum drawdown) but ask which *cross-sectional* risk metrics best forecast it.

Volatility persistence. That volatility is autocorrelated and forecastable is the founding fact of the ARCH/GARCH literature [1, 7, 11]. We ask whether this alone accounts for cross-sectional drawdown predictability.

Data snooping. Searching many signals inflates in-sample skill [14, 21]; the backtest-overfitting literature formalizes best-of-N inflation and prescribes deflation [4, 5]. Our contribution is to show an *autonomous* agent reaches this frontier at near-zero cost, so the holdout discipline becomes mandatory rather than advisable.

3 DATA AND PROTOCOL

Five test beds. *Crypto (survivorship-free):* a CoinMarketCap daily panel (2013–2018) that retains crashed and delisted coins, so a point-in-time top-100-by-market-cap universe includes assets that later crater; we use three regimes — 2016 (calm), 2017 (bull), 2018 (crash). *Equity:* S&P 500 constituents, 1994–1999 (calm) and 2005–2009 (GFC). The equity beds are survivorship-biased by construction — freely available equity data retains almost no true deaths, itself a telling finding — so the survivorship claim rests on the crypto beds, while equity tests cross-asset generality.

Skill and significance. From a 180-day trailing window we forecast the 90-day forward maximum drawdown of each asset (delisting scored as a −100% terminal loss); skill is the per-formation-date cross-sectional Spearman correlation between metric and realized drawdown, averaged over the bed, with paired block-bootstrap 95% confidence intervals. We use a *cross-sectional* rank correlation because the operational question is which assets to derisk on a given date, not the time series of any single asset. Formation dates are month-ends, so consecutive 90-day forward windows overlap; the block bootstrap over dates addresses the induced serial dependence, though on the 7–9-date crypto beds only coarsely. Because the evaluation spans ~45 metric × bed cells, significance flags are BH-corrected ($q = 0.05$) within each family. To ask what a metric adds *beyond* volatility, we use the partial Spearman correlation controlling for volatility. The discovery experiment (§8) uses a strict train/validation/*locked-test* split, scored once after the metric is frozen.

Pre-specified vs exploratory. The head-to-head verdict (§4), the magnitude-vs-shape partials (§5), and the equal-weight composite (§7) were specified before inspecting outcomes; the cross-bed heterogeneity and stress-conditional cuts (Appendices E, F) are exploratory and flagged as such. The crypto beds have only 7–9 formation dates, so their block-bootstrap CIs are coarse: we lean on the well-powered equity beds (54–66 dates) for the null that shape adds nothing, and read crypto-only positives as suggestive rather than confirmatory.

4 THE VERDICT: NOTHING BEATS VOLATILITY

Table 1 ranks each metric against volatility standalone. After BH correction, the only metrics that significantly beat volatility on *any* bed are two volatility variants (downside deviation, VaR) on the single crypto-bull bed — and they reverse on equity, where volatility significantly beats them. Every genuinely different, shape-based metric is significantly *worse* than volatility on every bed. There is no metric that consistently or generalizably beats volatility across regimes and asset classes. We fix that bar in advance to avoid a moving goalpost: an edge counts as *generalizable* only if the metric BH-significantly beats volatility on at least one crypto *and* one equity bed — i.e., survives a change of asset class. No metric clears it. This is a narrower and more defensible statement than "nothing beats volatility": small, regime-specific edges exist for volatility's own variants, but they do not survive a change of asset class.

Table 1: Standalone out-of-sample drawdown-forecast Spearman correlation. $*$ = significantly beats volatility, † = significantly worse (both survive BH, $q = 0.05$); unmarked = indistinguishable. The Hill tail exponent [15] is omitted here (not estimable per date on equity); its increment appears in Table 3.

Metric	cr-16	cr-17	cr-18	eq 94	eq 05
volatility (base)	0.62	0.36	0.38	0.61	0.56
downside deviation	0.62	0.38$*$	0.41	0.58†	0.54†
VaR	0.61	0.40$*$	0.36	0.58†	0.54†
ES [3]	0.62	0.35	0.33†	0.58†	0.52†
neg. semibeta [8]	0.40†	0.26†	0.31†	0.49†	0.46†
downside beta [2]	0.01†	0.09†	0.15†	0.24†	0.34†
lower-tail dep. [9]	−0.38†	−0.15†	−0.29†	−0.07†	−0.05†
GDA [12]	−0.06†	−0.11†	−0.05†	−0.24†	−0.32†
UPM/LPM [20]	−0.55†	−0.28†	−0.35†	−0.43†	−0.42†

5 THE ANATOMY: MAGNITUDE, NOT SHAPE

A verdict is not an explanation. If nothing beats volatility, what — if anything — is left to forecast beyond it, and which metrics capture it? Figure 1 plots the partial Spearman correlation $\rho(\text{metric, drawdown} \mid \text{volatility})$ for every metric and bed. The picture splits cleanly by *what a metric measures*, and the split survives BH correction. Tail-*magnitude* measures carry a small but genuine increment: VaR is significant on **all five beds** (partial ρ up to 0.19), downside deviation on the three crypto beds. Tail-*shape* measures — the Viole–Nawrocki asymmetry ratio, lower-tail dependence, the tail exponent, the disappointment-aversion proxy — are ≈ 0 or significantly *negative* everywhere.

A skeptic might counter that this is really an *own-asset vs systematic* split, not a magnitude vs shape one: the winners (VaR, ES, downside deviation) are all own-asset dispersion, whereas most losers (downside beta, semibeta, lower-tail dependence, the disappointment-aversion proxy) are *systematic* co-movement measures. Figure 1 is grouped to settle this. The magnitude/shape split survives with the own/systematic axis held fixed: the one *own-asset* shape metric, the Hill tail exponent, is itself ≈ 0 or negative on every bed — so among own-asset metrics, magnitude beats shape — and, separately, no systematic co-movement metric adds signal. Negative semibeta is the only shape statistic with an occasional positive increment, plausibly co-crash *magnitude* leaking into a shape measure. The forecastable residual beyond volatility is thus the *size* of tail losses, not their *asymmetry* — precisely the premise the shape metrics were built on. Nor is the increment an artifact of coding a delisting as a −100% loss: within the top-100 universe mid-window delistings are rare ($\leq 1\%$ of asset-dates), and valuing delisted assets at their last traded price, or dropping them entirely, leaves VaR's increment within ± 0.02 on every bed (Appendix G) — so this is crash-*severity* forecasting, distinct from the blow-up result of §9. Exact values, and the power checks that back the thin crypto beds, are in Appendix B.

This resolves an apparent paradox in Table 1: VaR is significantly *worse* than volatility standalone on equity, yet has a significant positive *increment* there (Figure 1). Both are true. As a standalone ranker VaR is a noisier proxy for the volatility it re-expresses, so

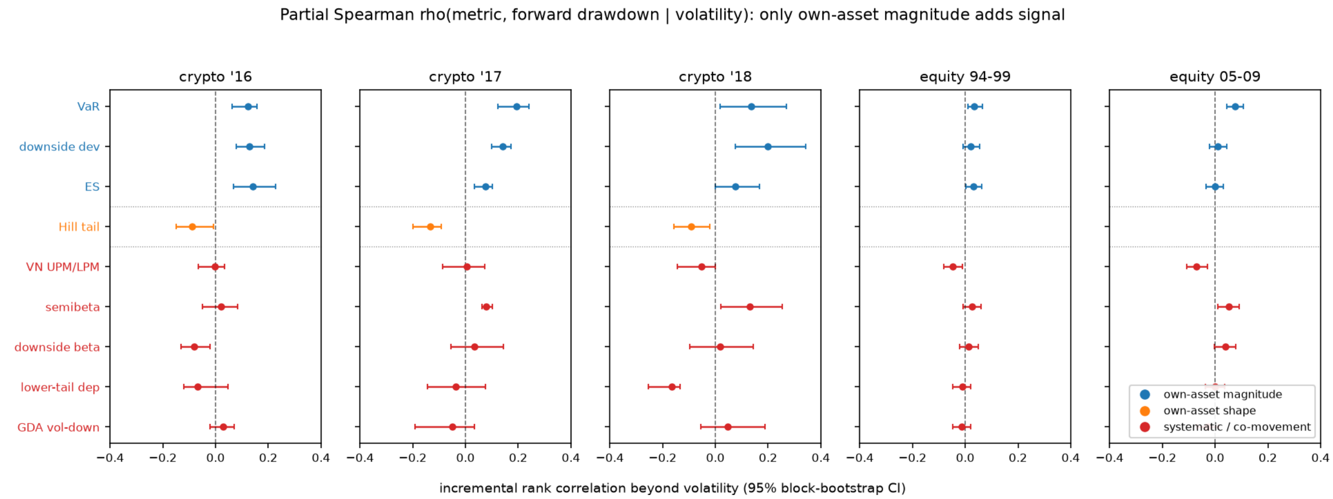

Figure 1: Incremental signal beyond volatility, by bed: partial Spearman $\rho(\text{metric}, \text{drawdown} \mid \text{volatility})$, grouped on both axes a skeptic cares about — what the metric measures (magnitude vs shape) *and* whose distribution it uses (own-asset vs systematic co-movement with the market). Only own-asset magnitude (blue) sits reliably right of zero. The lone own-asset *shape* metric, the Hill tail exponent (orange), is ≈ 0 or negative — so "magnitude beats shape" holds *within* the own-asset family — and no systematic co-movement metric (red) adds signal. Whiskers are 95% block-bootstrap CIs; dotted lines separate the three families.

it loses head to head; the partial correlation isolates its small orthogonal component. "Adds signal beyond volatility" and "beats volatility on its own" are different questions, and the decomposition separates them.

Why is even the magnitude increment larger in crypto? Because it scales with how fat-tailed and blow-up-prone a market is (Appendix E, Figure 6): the crypto beds (trailing excess kurtosis 8–21, blow-up rate 5–13%) show VaR increments of 0.12–0.19, versus 0.03–0.08 on the thin-tailed equity beds. Within crypto the increment further concentrates in high-volatility periods (Appendix F). Tail magnitude carries incremental information precisely when the tail is active.

6 WHY VOLATILITY IS HARD TO BEAT

Volatility wins because it is *persistent*, the founding fact of the ARCH/GARCH tradition [7, 11]. Trailing volatility forecasts forward volatility (rank correlation 0.42–0.84 across beds), and forward realized volatility is in turn strongly tied to forward drawdown (Appendix C, Table 4). This chain accounts for most of the forecastable variation.

It is tempting to go further and call forward volatility a *sufficient statistic* — to claim drawdown predictability simply *is* volatility persistence. A naive mediation test seems to confirm it: controlling for forward volatility, trailing volatility's partial correlation with drawdown collapses to ≈ 0 on equity. But that test is contaminated: forward volatility and forward drawdown are measured over the *same* window, so conditioning on one mechanically absorbs the other. When we measure the mediator on a *disjoint, earlier* window — volatility over days 1–30, drawdown over days 31–120 — so that it is genuinely *predictive*, the mediation vanishes: trailing volatility

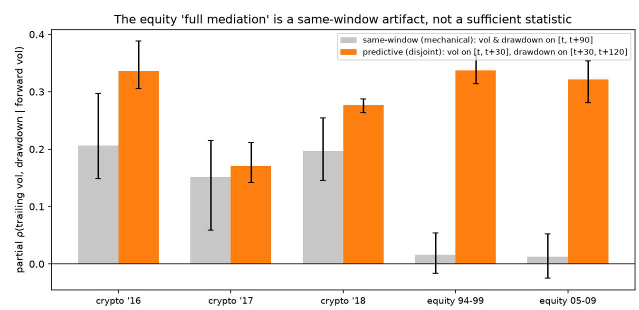

Figure 2: Mediation: partial $\rho(\text{trailing vol}, \text{drawdown} \mid \text{forward vol})$. Under the same-window (mechanical) design it is ≈ 0 on equity, falsely suggesting full mediation; under the predictive (disjoint-window) design it is 0.32–0.34 — trailing volatility is not screened off, so forward volatility is not a sufficient statistic.

retains a partial correlation of 0.32–0.34 on equity (Figure 2). Trailing volatility carries forecasting content a short-horizon volatility forecast does not. We therefore report volatility persistence as a *leading but incomplete* explanation — not a mechanism that reduces drawdown to a single number. (Consistent with this, drawdown skill and volatility persistence rise and fall together as the forecast horizon varies; Appendix D.)

7 DOES THE EDGE PAY? ECONOMICS AND A COMPOSITE

A rank correlation is not a profit-and-loss statement, and a partial correlation of 0.1 may or may not matter. We ask directly. First, the pre-registered composite the decomposition implies: $z(\text{vol}) + z(\text{VaR})$, equal weight, with no fitting (Appendix H). Its only fully out-of-sample test, a locked panel, shows *no significant gain* over volatility (Newey–West $t = 0.67$); across beds it beats volatility only on the fat-tailed crypto-bull bed and is significantly worse on thin-tailed equity, where a noisier VaR only degrades the volatility signal.

Second, a cost-aware backtest (Appendix **??**, Figure 7). At each date we sort the universe into risk terciles, hold the low-risk tercile, avoid the high-risk tercile, and measure realized drawdown net of a 20 bps turnover cost. The result is a genuine positive for volatility and a null for everything else: the high-risk tercile realizes 10–30 percentage points more forward drawdown than the low-risk tercile on every bed, and the composite adds nothing net of its higher turnover except, again, on crypto-bull. The economic picture matches the statistical one exactly: volatility is the workhorse; magnitude adds a little only where tails are fat.

8 CAN A MACHINE FIND MORE? AUTOMATED DISCOVERY

If the incremental signal beyond volatility is genuinely this small, then a tireless automated search should be *unable* to find a metric that beats volatility out of sample — and should instead overfit the validation set. That is exactly what happens, and it is worth demonstrating because AI agents are now being wired into quantitative-research loops. We let an LLM agent freely rewrite a scoring function, and separately ran a 20,000-configuration directed search. Both inflate the best *validation* score with search intensity while the *locked-test* score plateaus at the volatility baseline (Table 2, Figure 3). This plateau is the robust result: across a grid of 12 holdout panels × 3 search seeds (Appendix I), the validation-selected metric's out-of-sample edge over volatility is at most +0.07 and averages +0.02–0.03, straddling zero — no search, LLM or brute-force, finds a genuine out-of-sample improvement. A non-LLM random sweep reproduces it, so the danger is the cheap search, not the language model.

We are careful *not* to overstate a companion statistic. How often a validation-winner even *nominally* edges volatility on the test is regime-dependent, and far less informative than the near-zero size of that edge. On the hardest, most realistic holdout — validation in the 2017 bull market, test on the 2018 crash, where volatility's own cross-sectional skill *rises* (0.31 → 0.38) — validation-selected winners beat volatility only 43% of the time under the aggressive directed search (57% under a milder generator): a coin flip, despite a 0.94 validation–test rank correlation. On calmer out-of-time windows, where volatility weakens on the test, the nominal win-rate climbs toward 100% — but only by a noise-sized margin. Automated search's apparent success is largest exactly where it is least useful (calm futures) and evaporates where it would matter (stress futures).

The statistics are classical data snooping [5, 21]; what is new is the *setting*. An autonomous agent removes the last natural brake on the number of trials — analyst effort — and reaches the snooping frontier at near-zero marginal cost, with no human in the loop. The discipline that was merely advisable under human search becomes mandatory under agent search: freeze the scorer and data, score

Table 2: Best-of-N discovery. Validation inflates with search intensity; the locked test does not (volatility baseline 0.376). Validation and test are different splits, so compare trends, not levels.

Search intensity N	Best validation	Its locked test
1	0.27	0.31
100	0.35	0.37
20,000	0.356	0.373

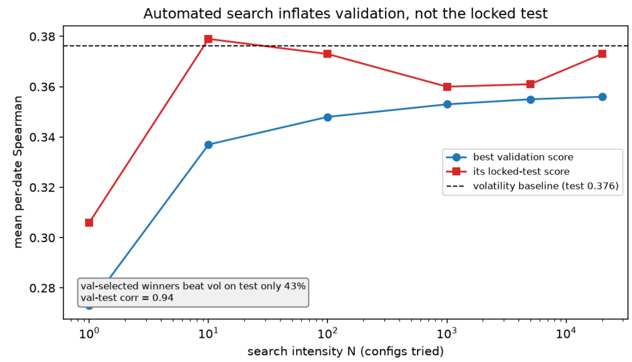

Figure 3: As search intensity grows, the best validation score climbs while its locked-test score stays flat around the volatility baseline. On this (2017 → 2018-crash) holdout, validation-selected winners beat volatility on the locked test only 43% of the time; across panels and seeds the win-rate is regime-dependent, but the locked-test score never exceeds the volatility baseline by more than noise (Appendix I).

a locked (ideally survivorship-free) holdout exactly once, log the trial count, and report the holdout win-rate alongside a deflated statistic.

9 WHAT VOLATILITY IS FOR: PREDICTING BLOW-UPS

The paper is not an argument against measuring risk — it is an argument for measuring it with volatility. The clearest positive use is forecasting outright blow-ups. On the locked crypto test, plain volatility predicts which assets crater (forward return ≤ −80%) with ROC-AUC 0.68 (0.95 at a −90% cutoff) and a 2.5× top-decile lift; a gradient-boosting model over all ten metrics *overfits and loses* to volatility alone (AUC 0.59). Consistent with §5, the size of recent moves — not their shape — is what carries the blow-up signal. This is a ranking result, not a deployment recommendation: turning it into action faces cost, liquidity, and reflexivity frictions we do not model, and in retail-heavy crypto markets such a signal raises information-asymmetry considerations beyond our scope.

10 DISCUSSION

The pieces form one coherent picture. Volatility sets a high bar because it is persistent (§6); the only thing left to add is the magnitude of tail losses, and only a little, mostly in fat-tailed markets (§5);

so a pre-registered composite yields no significant out-of-sample gain, no standalone metric beats volatility (§4), automated search cannot find one (§8), and the economics track the statistics (§7). The recurring failure of *asymmetry* and *shape* metrics — including the partial-moment ratio that motivated this study — is not an implementation accident: once symmetric volatility is accounted for, the asymmetric structure adds no incremental cross-sectional information for forecasting drawdown. We are deliberately careful about the *strength* of the persistence story: an earlier version of this project claimed forward volatility was a sufficient statistic, found that claim was a same-window measurement artifact, and retracted it here — just as an earlier "survivorship masks downside value" thesis was retracted when a hardened test refuted it. The paper reports only claims that survived stress-testing.

The practical takeaway is unglamorous but firm: for cross-sectional drawdown forecasting, use volatility — optionally with a small tail-magnitude term in fat-tailed markets — and treat any downside metric "discovered" to beat it, by hand or by machine, as unproven until it clears a locked, survivorship-free holdout.

11 CONCLUSION

On a survivorship-free, multi-regime, multiple-testing-corrected evaluation, no downside-risk metric — published or machine-discovered — reliably beats plain volatility at forecasting drawdown. What survives correction is a sharp decomposition: the only forecastable residual beyond volatility is tail *magnitude*, not *shape*. Volatility is hard to beat largely because it is persistent (a leading, though not sufficient, explanation), it is economically useful for separating drawdown and predicting blow-ups, and automated LLM-driven discovery cannot improve on it out of sample. Volatility is not glamorous, but on a level field it is very hard to beat.

Limitations. Skill is rank-correlation, not a full tradable P&L (§7 is a first economic check, not a portfolio study); the crypto beds have few formation dates (7–9), so their significance is suggestive and the null "shape adds nothing" leans on the well-powered equity beds; the repurposed pricing metrics are daily adaptations, and results speak to drawdown forecasting, not the originals' pricing claims.

A REPRODUCIBILITY

Code, data pointers, and result logs to reproduce every table and figure are available at https://github.com/meghanadhpulivarthi/downside-risk-metrics-study. Data sources: a CoinMarketCap daily crypto dump (retaining delisted coins), yfinance S&P 500 constituents and the market index, and point-in-time index membership.

B PARTIAL CORRELATIONS AND POWER

Table 3 gives the exact partial correlations plotted in Figure 1, with BH significance. VaR survives on all five beds; the one equity ES cell that is nominally significant does not survive correction. Power on the thin crypto beds: for VaR, on crypto-16 ($n = 8$) and crypto-17 ($n = 9$) the effect (0.12, 0.19) exceeds the 80%-power minimum detectable effect (0.09, 0.09) and is stable under leave-one-date-out; on crypto-18 ($n = 7$) the effect (0.13) is below its MDE (0.18) and is reported as suggestive. A placebo (permuting forward drawdown within each date) drives the trailing-volatility correlation to ≈ 0 on four of five beds; the exception, crypto-18 (+0.028, CI $[0.003, 0.059]$), is negligible against a real skill of ≈ 0.38 and reflects the coarse block bootstrap at $n = 7$.

Table 3: Partial Spearman ρ(metric, drawdown | volatility). Rows 1–3 tail magnitude, rows 4–9 tail shape. $*$ survives BH ($q = 0.05$, positive); † survives BH negative; n/e = not estimable per date.

Metric	cr-16	cr-17	cr-18	eq 94	eq 05
VaR	+0.12$*$	+0.19$*$	+0.13$*$	+0.03$*$	+0.08$*$
downside dev.	+0.13$*$	+0.14$*$	+0.20$*$	+0.02	+0.01
ES [3]	+0.14$*$	+0.08$*$	+0.08	+0.03	+0.00
neg. semibeta [8]	+0.02	+0.08$*$	+0.13$*$	+0.03	+0.05$*$
UPM/LPM [20]	−0.00	+0.01	−0.05	−0.05†	−0.07†
downside beta [2]	−0.08†	+0.03	+0.02	+0.01	+0.04
Hill tail [15]	−0.09†	−0.13†	−0.09†	n/e	n/e
lower-tail dep. [9]	−0.07	−0.04	−0.17†	−0.01	+0.00
GDA [12]	+0.03	−0.05	+0.05	−0.01	−0.03

Table 4: Persistence chain (mean per-date Spearman; all CIs exclude 0).

Bed	trail→fwd vol	fwd vol→dd	(trail→dd)
crypto-16	0.68	0.77	0.62
crypto-17	0.42	0.62	0.36
crypto-18	0.42	0.53	0.38
equity 94	0.84	0.71	0.61
equity 05	0.80	0.70	0.56

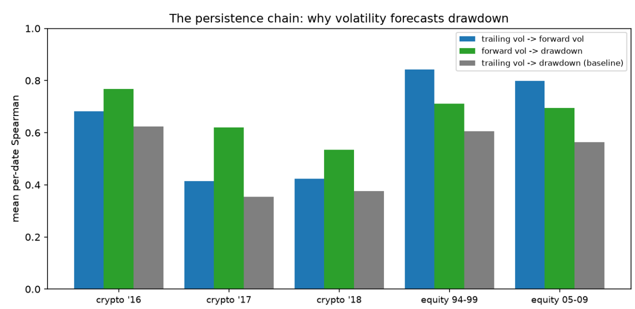

Figure 4: The two links of the persistence chain relative to the direct trailing-volatility→drawdown skill, by bed.

C THE PERSISTENCE CHAIN

Table 4 decomposes why trailing volatility forecasts drawdown. Both links exceed the direct trailing-vol→drawdown skill they compose. The forward-vol→drawdown column is contemporaneous (both over the same forward window) and hence partly mechanical [1]; the genuinely predictive link is trailing→forward volatility (see §6 and Figure 4).

D HORIZON DYNAMICS

Sweeping the forecast horizon from 14 to 90 days, volatility persistence and drawdown skill move together (Figure 5): on equity both rise (persistence 0.66 $\rightarrow$ 0.84, skill 0.47 $\rightarrow$ 0.61, at a roughly constant gap); on crypto both are flat. Wherever persistence is higher, so is skill — consistent with persistence driving most of the forecastable variation, with the constant gap the part it does not explain.

E CROSS-BED HETEROGENEITY

Figure 6 relates the VaR increment to each bed's tail fatness. The increment is systematically larger in the fat-tailed, blow-up-prone crypto beds than in equity; with only five beds this is descriptive, but the direction is consistent with the conditional evidence in Appendix F.

F CONDITIONAL INCREMENT (STRESS VS. CALM)

Pooling dates by asset class and splitting each pool at its median aggregate (cross-sectional-median) volatility, the VaR increment concentrates in high-stress dates within crypto — rising from +0.13 (CI $[0.06, 0.18]$) in low-stress dates to +0.18 ($[0.09, 0.24]$) in high-stress dates (24 dates, 12 per regime; suggestive given the overlap) — while on equity it is flat and small (+0.056 vs. +0.050; 120 dates). Tail magnitude carries incremental information precisely when the tail is active.

G DELISTING-CODING ROBUSTNESS

The crypto forward-drawdown codes a mid-window delisting as a terminal −100% loss. To confirm the magnitude increment is genuine crash-severity skill — not an artifact of that code, nor a restatement of the blow-up result (§9) — we recompute $\rho(\text{metric}, \text{drawdown} \mid \text{vol})$ under two alternative codings: valuing a delisted asset at its *last traded price* (ordinary peak-to-trough, no −100% override), and *excluding* delisted assets (skill among survivors only). Within the point-in-time top-100 universe, mid-window delistings are rare — 0.9% of asset-dates on crypto-2016 and none on the other four beds — so the three codings barely differ. VaR's increment is unchanged to within ±0.02: per bed, baseline / last-price / exclude gives +0.12/+0.12/+0.14 (crypto-16), +0.19 on all three (crypto-17), +0.14 on all three (crypto-18), +0.035 (equity-94), and +0.076 (equity-05); downside deviation and ES behave identically. The magnitude edge is drawdown-*severity* forecasting, distinct from the terminal-death signal.

H PRE-REGISTERED COMPOSITE AND ECONOMIC BACKTEST

The composite is $z(\text{vol}) + z(\text{VaR})$, equal weight, computed cross-sectionally per date, with no fitting. VaR was chosen a priori as the magnitude metric surviving correction on all five beds (§5); the weights are untuned, so the fully out-of-sample evidence is the locked-panel test (Table 5), which is not significant. The economic backtest (§7, Figure 7) sorts the universe into risk terciles and measures realized-drawdown discrimination (forward DD of the high- minus low-risk tercile) and the forward-return spread, net of a 20 bps turnover cost.

Table 5: Composite $z(\text{vol}) + z(\text{VaR})$ vs. volatility: per-date Spearman and difference (95% CI). Significant only on crypto-bull; locked-panel test not significant.

Test bed	vol ρ	comp ρ	difference (95% CI)
crypto-16	0.62	0.62	−0.00 (n.s.)
crypto-17	0.36	0.40	+0.05 $[0.02, 0.07]$
crypto-18	0.38	0.39	+0.01 (n.s.)
equity 94	0.61	0.60	−0.008 $[-0.011, -0.004]$
equity 05	0.56	0.56	−0.00 (n.s.)
Locked panel: 0.376 $\rightarrow$ 0.388, Newey–West $t = 0.67$ (n.s.).			

I SNOOPING ROBUSTNESS ACROSS PANELS AND SEEDS

To check that the discovery result is not a single-panel artifact, we re-ran the candidate search (a general nonlinear generator over per-date feature ranks, 1,500 candidates per run) over a grid of holdout panels × 3 search seeds, in two families: *out-of-time* — a fixed-length test window slid through the timeline with validation always chronologically earlier (six windows, including the original 2017 $\rightarrow$ 2018 split) — and *random* — order-ignoring 10-val/7-test repartitions of the non-train dates (six draws). Per run we record the validation-selected metric's locked-test edge over volatility (*test gap*) and the fraction of validation-winners that also beat volatility on test (*win-rate*).

The plateau is stable. The test gap averages +0.020 (out-of-time; range −0.028 to +0.046) and +0.035 (random; +0.014 to +0.072) — always small and often straddling zero, so no search beats volatility out-of-sample by more than noise on any panel or seed. Validation–test rank correlation is 0.94–0.99 throughout. *The win-rate is not.* It averages 0.79 (out-of-time, range 0.55–1.00) and 0.82 (random, 0.62–0.97), lowest on holdouts where volatility's own skill is high on the test (the 2018 crash: 0.43 directed / 0.57 milder generator) and near 1.0 on calm windows. Because these nominal "wins" come with a test gap of essentially zero, the *sign* of the win-rate is far less informative than its *magnitude*: the validation-selected metric matches, but does not beat, volatility.

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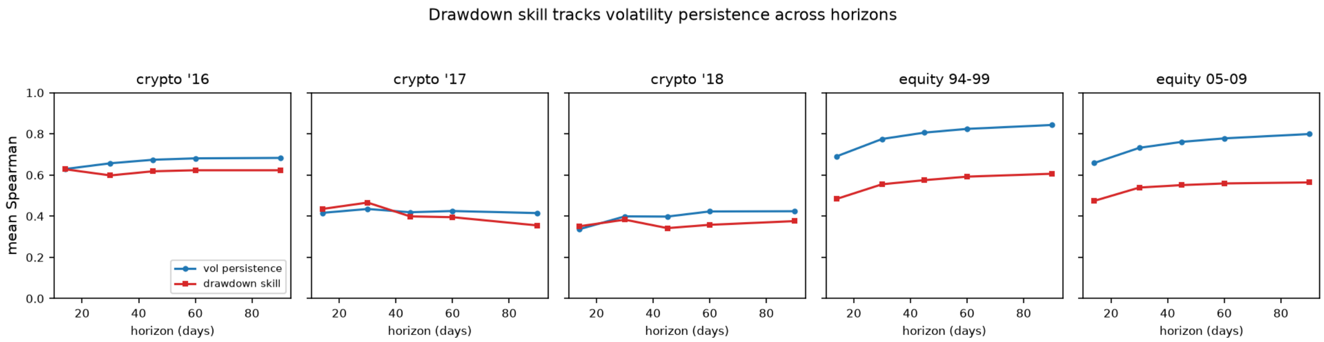

Figure 5: Volatility persistence (blue) and drawdown-forecast skill (red) versus forecast horizon, by bed. The curves track.

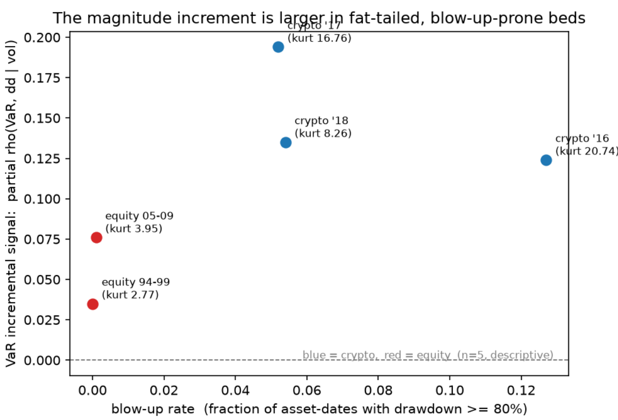

Figure 6: VaR incremental signal (partial ρ | volatility) versus blow-up rate, by bed. Larger in fat-tailed crypto; descriptive ($n = 5$).

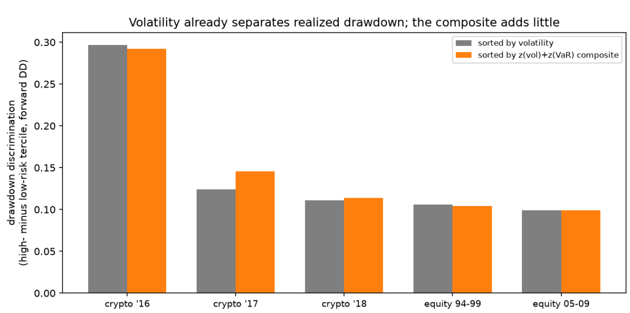

Figure 7: Realized drawdown discrimination (high- minus low-risk tercile) by bed, sorting on volatility vs. the composite. Volatility already separates drawdown strongly; the composite is indistinguishable except marginally on crypto-bull.

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